BHUVAN M ACHARYA

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Education

Indian Institute of Information Technology Tiruchirappalli — Tiruchirappalli, India
B.Tech. in Computer Science and Engineering | CGPA: 8.5/10 — *2024 – 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Machine Learning

Experience

Nebula Technologies Labs Pvt. Ltd. — Remote
Founding Software Engineer / Associate Software Engineer Intern — *May 2026 – Aug. 2026*

- Architected and developed a production CRM platform using **Java, Spring Boot, React, and PostgreSQL**, supporting **100+ active users** across **20+ business workflows**.
- Designed production infrastructure using **Docker, Nginx, VPS hosting, and Cloudflare**, achieving **99.9% uptime** and a **30% reduction in production incidents**.
- Built automated **CI/CD pipelines using Jenkins and GitHub Actions**, reducing deployment time from approximately **10 minutes to under 1 minute** and saving 5 engineering hours monthly.
- Optimized PostgreSQL operations through indexing, query optimization, transactions, isolation levels, migrations, JSONB, and connection pooling, contributing to a **25% reduction in query latency**.
- Developed concurrent and asynchronous backend components using **Java multithreading, ExecutorService, thread pools, concurrent collections, RabbitMQ, and Kafka**.
- Implemented and tested backend services using **JUnit, Postman, and Bruno**; debugged production issues involving APIs, database failures, backend exceptions, and deployment workflows.
- Collaborated in a **6-member engineering team** across system architecture, technical design, code reviews, Jira-based issue tracking, debugging, release workflows, and production deployments.

Indian Institute of Information Technology Tiruchirappalli — Tiruchirappalli, India
Undergraduate Research Collaborator — *2026 – Present*

- Developing an open-source graph-native platform for reproducible computational research, focusing on dataset versioning, experiment tracking, artifact management, and workflow orchestration.

Projects

CampusHire — Placement CRM & ERP Platform — *Java / Spring Boot / React / PostgreSQL*

- Developed a full-stack enterprise platform with **REST APIs, configurable workflows, role-based access control, secure authentication, and modular backend services**.
- Architected a metadata-driven backend supporting dynamic forms, records, layouts, relationships, and configurable business processes using PostgreSQL and JSONB.
- Designed backend components for authentication, authorization, database architecture, asynchronous processing, caching, and workflow orchestration.
- Containerized services and automated deployment using **Docker, Jenkins, and Nginx**.

NeuroWeave — Graph-Native Research Platform — *Python / React / PostgreSQL*

- Building an open-source platform for reproducible computational research using graph-based workflow management and experiment lineage.
- Developing experiment tracking, artifact versioning, dataset management, and workflow orchestration capabilities for scientific computing environments.

Technical Skills

Languages: Java, C++, Python, Go, JavaScript, TypeScript, SQL, C

AI/ML & Deep Learning: Machine Learning, Deep Learning, Neural Networks, Natural Language Processing (NLP), CNNs, Feature Engineering, Model Training & Evaluation, Hugging Face

LLM Engineering & Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, LangGraph, Ollama, Prompt Engineering, MCP, Tool Calling, Structured Outputs, AI Agents, Agentic Workflows, LLM Evaluation

AI-Assisted Development: Cursor, GitHub Copilot, Claude Code, OpenCode, AI Coding Agents, Vibe Coding, Context Engineering, Prompt Engineering, AI-Assisted Debugging, Code Generation, Automated Testing, AI Code Review, Rapid Prototyping

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks, Concurrency, Distributed Systems

Backend & System Design: Spring Boot, Node.js, Express.js, REST APIs, gRPC, Microservices, Event-Driven Architecture, Asynchronous Processing, Workflow Orchestration

Databases: PostgreSQL, MySQL, Redis, Vector Databases, Indexing, Query Optimization, Transactions, ACID, Isolation Levels, JSONB, Database Design

Concurrency & Messaging: Java Multithreading, ExecutorService, Thread Pools, Locks, Concurrent Collections, RabbitMQ, Kafka, Message Queues

Cloud & DevOps: Docker, Kubernetes, Jenkins, GitHub Actions, Nginx, AWS, Linux, VPS, Cloudflare, CI/CD

Testing & Engineering Tools: JUnit, Postman, Bruno, Git, GitHub, Debugging, Code Reviews, Jira, System Design

Achievements & Open Source

Competitive Programming: Solved **300+ problems on LeetCode**; achieved a maximum **Codeforces rating of 1180**; participated in college programming contests.

Open Source: Active open-source contributor with public GitHub projects and merged pull requests; applied twice to Google Summer of Code and actively contributes to open-source software.

Leadership & Campus Involvement

Training & Placement Representative IIIT Tiruchirappalli
Student Placement Representative *2025 – Present*

- Coordinated recruiter, student, and placement operations while contributing to the development of the institution's placement CRM/ERP platform.

Head — Web Development Club IIIT Tiruchirappalli
Web Development Lead *2025 – Present*

- Mentored student developers and contributed to maintaining institutional web applications and college websites.

Co-Founder & Head — Quantum Club IIIT Tiruchirappalli
Founding Member and Club Lead *2026 – Present*

- Co-founded the institution's first Quantum Club alongside a fellow student, establishing a student-led community focused on quantum computing education, technical sessions, and hands-on workshops.

Certifications

Deep Learning Specialization — DeepLearning.AI 2025–2026

Machine Learning Specialization — Stanford University & DeepLearning.AI 2025

Natural Language Processing — DeepLearning.AI 2026